

Assignment 2: TCP Wireshark Performance mininet

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2..Mininet Topology

Bandwidth = 5 Mbps

Delay = 50 ms

h1 (Server)

h2 (Client)

IP: 10.0.0.1

IP: 10.0.0.2

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+----- s1 -----+

3..Explanation of TCP Client-Server Communication

TCP client-server communication is a process where a **client establishes a reliable connection with a server using a TCP three-way handshake** (SYN, SYN-ACK, ACK). After the connection is established, the client sends messages to the server, and the server processes the messages and sends acknowledgment (ACK) responses.

In this experiment, **h2 acts as the client and h1 acts as the server**. The client sends messages in the format MSG_ID | MESSAGE_SIZE | MESSAGE_DATA, and the server replies with ACK | MSG_ID | RECEIVED_SIZE. After all communication is completed, the TCP connection is closed using the connection termination process (FIN and ACK packets).

4..Difference between Persistent and New-Connection Modes

Persistent connection mode keeps a single TCP connection open and uses it to send multiple messages. It requires only one TCP handshake and one connection close, resulting in lower response time, less overhead, and better throughput.

New-connection mode creates a separate TCP connection for each message. Every message requires a new TCP handshake and connection termination, which increases delay and overhead, leading to higher response time.

Feature	Persistent Mode	New-Connection Mode
TCP connection	One connection for all messages	New connection for each message
TCP handshakes	One handshake	Multiple handshakes
Connection closing	One close at the end	Close after every message

Response time	Lower	Higher
Overhead	Less	More
Throughput	Better	Lower
Example	Chat applications, streaming	Short independent requests like attendance submission

8..Questions:

1. Why does TCP use a three-way handshake?

TCP uses a three-way handshake to establish a reliable connection between client and server. It allows both sides to synchronize sequence numbers and confirm that both devices are ready to communicate. The steps are SYN, SYN-ACK, and ACK.

2. Which mode had lower response time?

Persistent connection mode had lower response time because the TCP connection is established only once and reused for all messages. It avoids the repeated connection setup and closing overhead.

3. Why does new-connection mode have more overhead?

New-connection mode creates a separate TCP connection for every message. Each message requires a new handshake and connection termination, which increases packet exchange, delay, and overall overhead.

4. Which mode is better for CampusChat and why?

Persistent connection mode is better for CampusChat because users continuously send and receive messages. Keeping one connection open provides faster communication and reduces connection overhead.

5. Which mode is suitable for NetAttend and why?

New-connection mode is suitable for NetAttend because attendance requests are short and independent. Creating a connection only when needed saves resources when there is no continuous communication.

6. How does message size affect throughput?

As message size increases, throughput generally increases because more data is transmitted per connection. Larger messages reduce the effect of TCP overhead compared to smaller messages.

7. How did Wireshark help verify TCP behavior?

Wireshark helped by capturing TCP packets and showing the handshake, data transfer, and connection termination. It verified the difference between persistent mode and new-connection mode by displaying the number of TCP connections created.